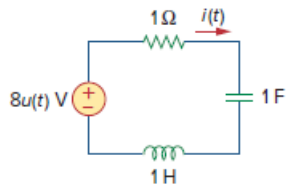


Problem

Determine $i(t)$ in the circuit of Fig. 16.35 by means of the Laplace transform.

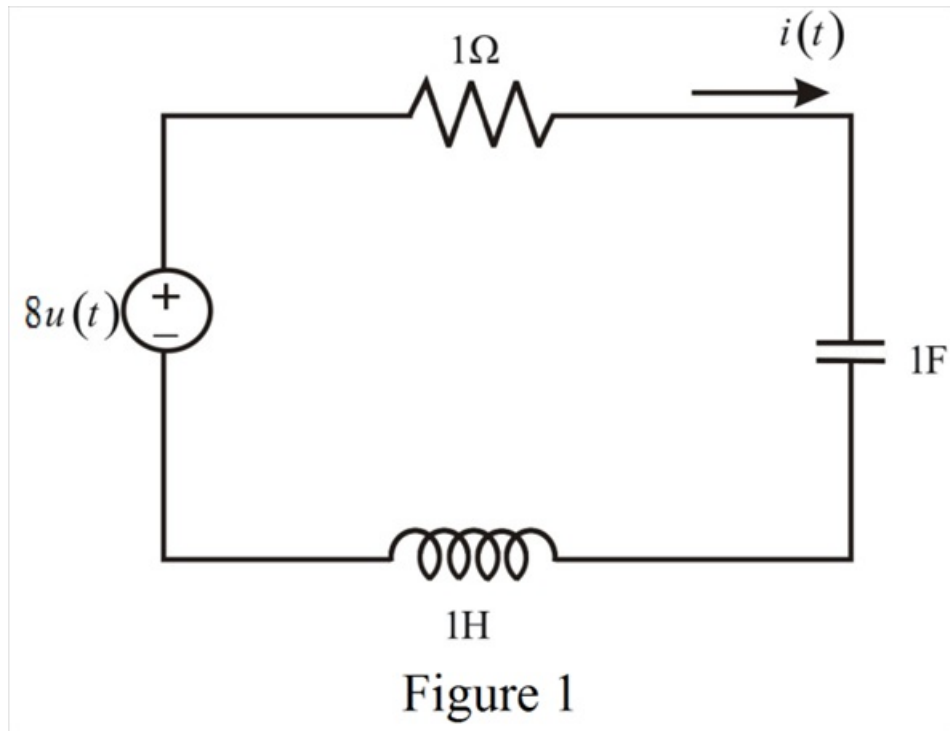
Figure 16.35:



Step-by-step solution

Step 1 of 5

The following is the circuit diagram:



Step 2 of 5

Determine $i(t)$ in the circuit in Figure 1.

Use Laplace transform and transform the circuit from time domain to s domain.

$$L[u(t)] = \frac{1}{s}$$

$$L[i(t)] = I(s)$$

The Laplace equivalent of 1 H is

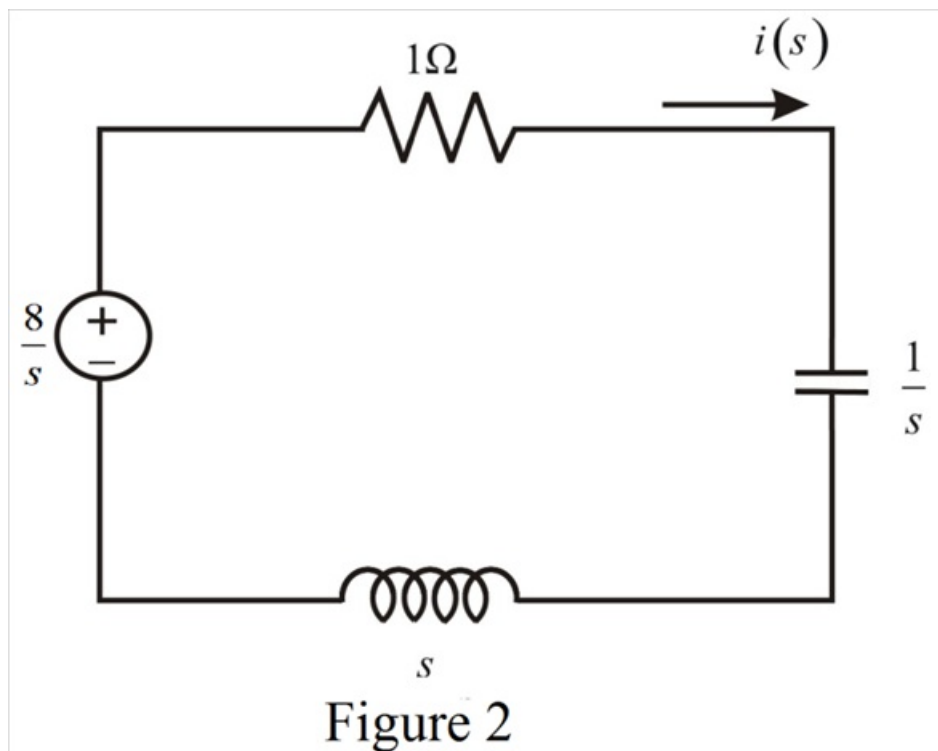
$$sL = s(1) \\ = s$$

The Laplace equivalent of 1 F is

$$\frac{1}{sC} = \frac{1}{s(1)} \\ = \frac{1}{s}$$

Step 3 of 5

The s -domain equivalent circuit for the circuit in Figure 1 is shown in Figure 2.



Step 4 of 5

Apply Kirchhoff's Voltage Law to the circuit in Figure 2.

$$\frac{8}{s} = \left(1 + s + \frac{1}{s}\right) i(s)$$

$$\frac{8}{s} = \left(\frac{s + s^2 + 1}{s}\right) i(s)$$

$$i(s) = \frac{8}{s^2 + s + 1}$$

Re-write $i(s)$ as:

$$\begin{aligned} i(s) &= \frac{8}{s^2 + 2 \times \frac{1}{2} \times s + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1} \\ &= \frac{8}{\left(s + \frac{1}{2}\right)^2 + \frac{3}{4}} \\ &= \frac{8}{\left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \frac{2}{\sqrt{3}} \left(\frac{\frac{\sqrt{3}}{2} (8)}{\left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \right) \end{aligned}$$

Hence,

$$i(s) = \frac{2}{\sqrt{3}} \left(\frac{\frac{\sqrt{3}}{2} (8)}{\left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \right) \dots\dots (1)$$

Recollect the Laplace inverse of $\frac{a}{(s+b)^2+a^2}$.

$$L^{-1}\left[\frac{a}{(s+b)^2+a^2}\right] = e^{-bt} \sin at$$

Hence,

$$L^{-1}\left[\frac{\frac{\sqrt{3}}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right] = e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

Write the equation for $i(s)$.

$$i(s) = \frac{2}{\sqrt{3}} \left(\frac{\frac{\sqrt{3}}{2}(8)}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} \right)$$

Apply inverse Laplace transform on both sides of equation (1).

$$i(t) = \frac{16}{\sqrt{3}} \left(e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right) \right)$$

Thus, the current in the circuit is, $i(t) = 9.237 \left(e^{-0.5t} \sin(0.866t) \right) \text{ A}$.